

EXERCISES

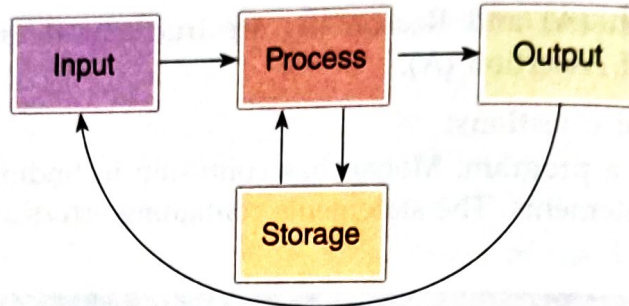


PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions

Tick (✓) the correct answer:

1.



What does the above illustration depict?

- (a) POP cycle (b) IPO cycle (c) OOP cycle (d) IOP cycle
2. The Scanner class method used to accept words with space:
(a) next() (b) nextLine() (c) Next() (d) nextString()
[ICSE 2012]
3. The keyword used to call package in the program:
(a) extends (b) export (c) import (d) package
[ICSE 2012]
4. Which of the following classes is a member of java.io package?
(a) InputStreamReader class (b) Scanner class
(c) cardLayout class (d) All of them
5. Missing a semicolon in a statement is what type of error?
(a) Logical error (b) Syntax error (c) Runtime error (d) No error
[ICSE 2012]
6. The style of expressing single line comment is:
(a) /*comment*/ (b) * comment
(c) //comment (d) /* comment
7. What is default delimiter used as token separator using Scanner class?
(a) Comma (b) Whitespace (c) semicolon (d) colon
8. BufferedReader class is used in association with:
(a) Scanner class (b) Wrapper class
(c) InputStreamReader class (d) None of these
9. IPO stands for:
(a) Inter Processing Operator (b) Input Process Output
(c) Internal Parity Operator (d) Internal Processing Option
10. Read the following text and choose the correct answer:
It is a non-executable statement written in a program that enables the users to understand the program logic.
What is described in the above statement?
(a) Remark (b) Comment (c) Method (d) Execution

II. Name the following:

1. a package needed to import the Scanner class
2. a method that accepts a character through Scanner object
3. a package needed to import InputStreamReader class
4. a method to accept a real number through Scanner object
5. a method that accepts an integer token through Scanner object

III. Assertion and Reasoning based questions:

Assertion (A): Given: float p= in.nextFloat();

Reason (R): It means that the variable p will store a float type as well double type value through the Scanner object.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

IV. Application based questions:

As a homework assignment, your teacher has given a program to swap the values of two memory locations. In the program, she has left some places marked blank with ?1?, ?2?, ?3? and ?4? to be filled with appropriate value/ expression. The incomplete program is as given below:

```
import java.....?1?.....
class Swap
{
    public static void main(String args[ ])
    {
        Scanner sc = .....?2?..... Scanner(System.in);
        int n = 5;
        int m = 8;
        int p = .....?3?.....
        n = m;
        m = .....?4?.....;
        System.out.println(n+"\t"+m);
    }
}
```

With reference to the above program, answer the following questions:

- (a) Name the package to be filled in the place marked with ?1?
- (b) Which keyword will be filled in place of ?2? that is used for dynamic allocation of Scanner object?
- (c) What variable should be filled in place marked with ?3?
- (d) Name the value to be filled in the place marked with ?4?

V. Write down the syntax with reference to Java Programming:

1. to accept an integral value 'p' through InputStreamReader class
2. to accept a fractional value (float) 'm' through Scanner class
3. to accept a character 'd' through InputStreamReader class
4. to accept a fractional value 'n' in double data type through InputStreamReader class
5. to accept a word 'wd' through InputStreamReader class
6. to create a Scanner object

VI. Unsolved Java Programs:

1. The time period of a simple pendulum is given by the formula:

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Write a program to calculate and display the time period of a simple pendulum

by taking length (l) and acceleration due to gravity (g) as inputs. [$\pi = \frac{22}{7}$]

2. Write a program by using class 'Employee' to accept Basic Pay of an employee. Calculate the allowances/deductions as given below.

Allowance/Deduction Rate

Dearness Allowance (DA) : 30% of Basic Pay

House Rent Allowance (HRA) : 15% of Basic Pay

Provident Fund (PF) : 12.5% of Basic Pay

Gross Pay = Basic Pay + Dearness Allowance + House Rent Allowance

Net Pay = Gross Pay – Provident Fund

Finally, find and display the Gross and Net pay.

[ICSE-2005]

3. A shopkeeper offers 10% discount on the printed price of a digital camera. However, a customer has to pay 6% GST on the remaining amount. Write a program in Java to calculate and display the amount to be paid by the customer taking printed price as an input.
4. A shopkeeper offers 30% discount on purchasing an article whereas the other shopkeeper offers two successive discounts 20% and 10% for purchasing the same article. Write a program in Java to compute and display the discounts. Take the price of an article as the input.
5. Write a program to input the time in seconds. Display the time after converting it into hours, minutes and seconds.
Sample Input: Time in seconds 5420
Sample Output: 1 Hour 30 Minutes 20 Seconds
6. Write a program to input two unequal numbers. Display the numbers after swapping their values in the variables without using a third variable.
Sample Input: a = 23, b = 56
Sample Output: a = 56, b = 23
7. A certain sum is invested at the rate of 10% per annum for 3 years. Write a program to find and display the difference between Compound Interest (CI) and Simple Interest (SI). Take the sum as an input.

[Hint: $SI = \frac{P * R * T}{100}$

$A = P*(1+R/100)^T$ and $CI = A - P$]

8. A shopkeeper sells two calculators for the same price. He earns 20% profit on one and suffers a loss of 20% on the other. Write a program to find and display the total cost price of the calculators by taking their selling price as input.

$$\left[\text{Hint: } CP = \left(\frac{SP}{1 + \frac{\text{profit}}{100}} \right) \text{ (when profit), } CP = \left(\frac{SP}{1 - \frac{\text{loss}}{100}} \right) \text{ (when loss)} \right]$$

PART B – SUBJECTIVE QUESTIONS

I. Differentiate between the following:

1. nextInt() and nextFloat() methods
2. Syntax and logical errors
3. Testing and Debugging

II. Answer the following:

1. What is meant by comment? Name the various ways to give comment in Java programming.
2. What do you mean by the Scanner class?
3. What are the different ways to give inputs in a Java Program?
4. What is the use of the keyword 'import'? [ICSE 2010]
5. What is a package? Give an example. [ICSE 2006]
6. Give an example to input an integer number by using function argument.
7. "Diagnostical messages detect only syntax errors". Comment on the statement.
8. Write down the syntax to input a character through the Scanner class with an example.
9. What do you understand by 'Run Time' error? Explain with an example.
10. What are the different types of errors that take place during the execution of a program?